26-April-2026

Agenda:

FE-3

Imputation technique:

→ Simple imputation
→ KNN
→ Regression Imputation (Advance)

ML
↳ Linear Regression
↳ KNN

x	y
1	2
2	4
3	6
4	8
5	10
6	12

→ $y = 2x$

$y = mx + c$
$y = wx + b$ } straight line

$x \longrightarrow$
$y \longrightarrow$ ML → $y = mx$ → m?

$1 \times 1.9 \quad 1.9 \longleftrightarrow 1.9$

$2 \times 1.9 \quad 3.9 < 3.8$

$3 \times 1.9 \quad 4.8 \quad 5.7$

$m \rightarrow ?$

1.5

$1 \times 1.5 \quad 1.5 \quad 1.9$ (0.4)

$2 \times 1.5 \quad 3.0 \quad 3.4$ (0.4)

$3 \times 1.5 \quad 4.5 \quad 4.8$

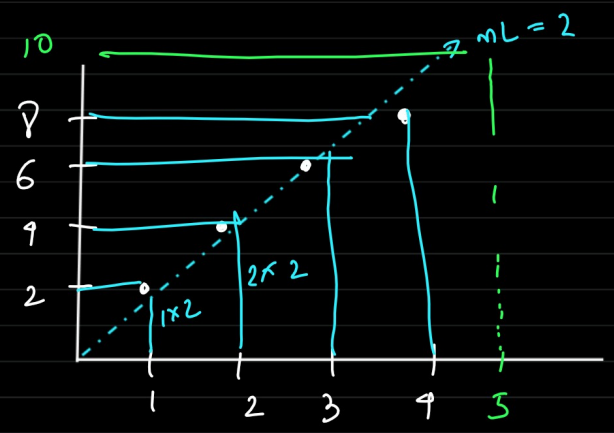
0.3 → error

error ← pred − Actual

$x, y \rightarrow ML \rightarrow m? \rightarrow$

↳ error

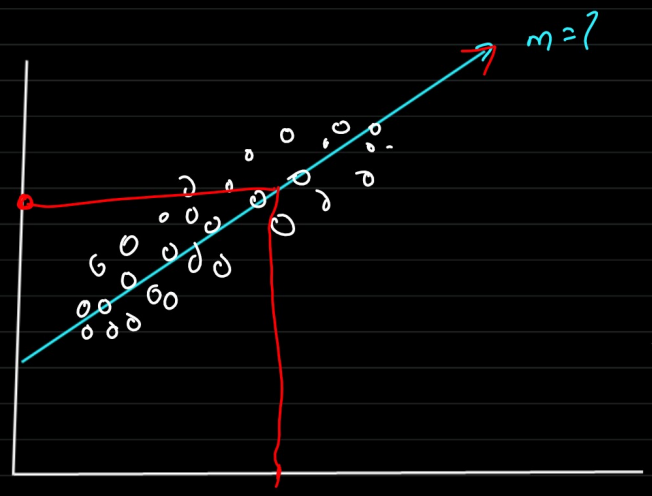
x	y
1	2
2	4
3	6
4	8



$x, y \rightarrow ML \rightarrow y = mx$

↓

2



kaggle dataset

↓

numerical column

↓

missing data

↓

data without NA ← total data − missing data

→ good data → ML

↓

missing → learn → $\boxed{}$

$x_1 \quad x_2 \quad x_3 \quad x_4 \ldots \quad x_N \qquad y$

(No null) Fill → y

Data → Train, Test

Train → ML → learns

Data → predict → Test (Null)

Train → $x_1 \quad x_2 \quad x_3 \quad x_4 \ldots x_{66}$

$x_1 \ldots x_{65} \rightarrow X$

Lotfrontage → y

→ ML → <u>option - 1</u>

$x_1, x_2, x_3 \rightarrow X$

Lotfrontage → y

→ ML → option - 2

Train → Null → ML

[ ] →

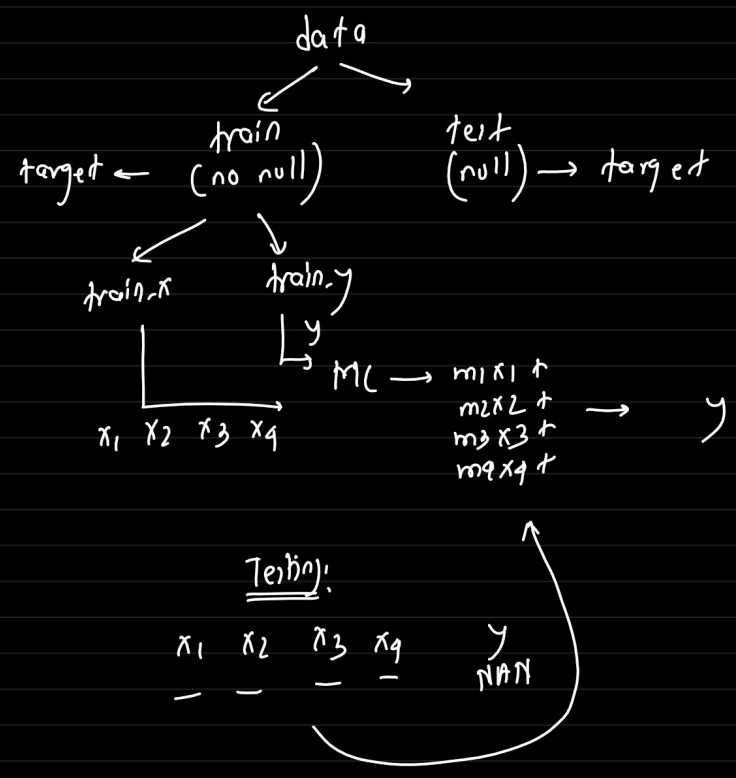
X-train → ML → $m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4$ →

Y-train

$x_1 \quad x_2 \quad \boxed{x_3} \quad x_4 \quad y$ → Trained (ML) → $m_1 x_1 + m_2 x_2 + m_3 x_3 + m_4 x_4$ → y

[ ] - - - ( )

<u>sol.</u>



$x_1 \quad x_2 \quad x_3 \quad x_4 \longrightarrow \boxed{y}$

function → numerical → null values → [ ]

for i in [ ]:

corr[i].abs() vs other → >0.1 [ ] < 0.5

colums → [ ]

train_data → colums > 30% → reject

→ mean, mode

↳ rej [ ]

↳ [ ]

FE:

(1) Handling missing data

→ remove columns

→ Imputing missing value

(2) Feature selection

(3) Handling outliers

Handling outliers :—

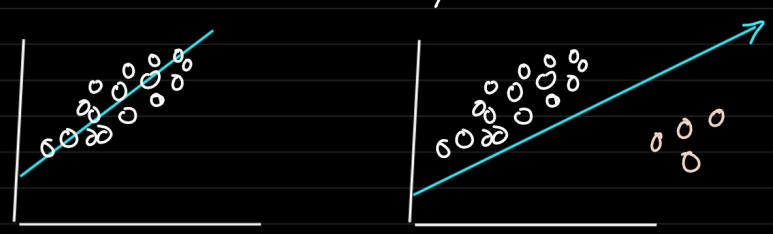
outlier - extreme value

students → Majority → 150 - 180 cm

↳ 260 cm → outliers

why care about outliers

→ ML models like linear Regression is sensitive to outliers.



→ Distance based models (KNN) get skewed by outliers

→ outliers represent error or rare but important cases

Method to detect & Handle outliers :

1: visualization (Box plot)(scatter)

2: IQR method → (range)

3: z-score → 3

4: (capping) the extreme value → winsorization

± 3 99.7

Hypothesis testing : ( ) function

H_0 → Not normal

H_1 → nor

data

z-score

p-value

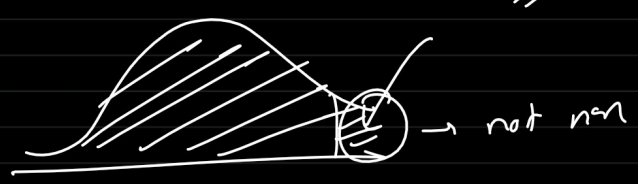
H_0 ←

H_1 ←

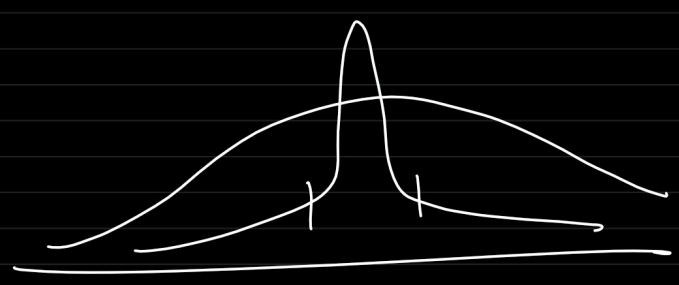
z-score

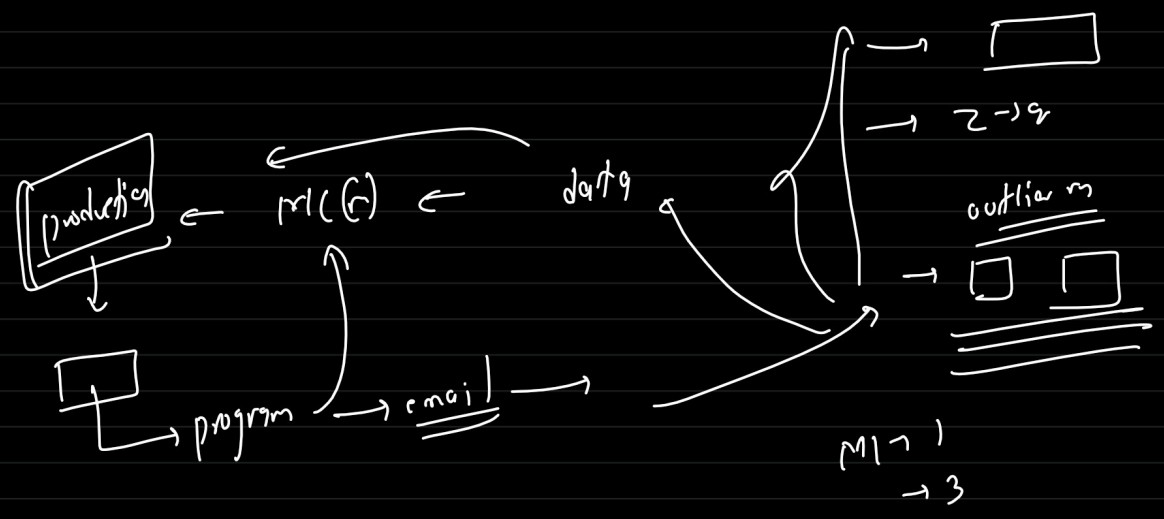
p-val

95%



→ not nor




production
ML(G)
data
2->9
outliers
program
email
ML->1
->3